

REPORT

Section A

- a) No if two variables exhibit a strong correlation with a third variable does not necessarily mean that they will also display a high degree of correlation with each other. For eg- If the mileage of the car depends on both the fuel type and the type of tires , the mileage exhibits a strong relation with fuel type as better quality fuel will give better mileage and also better quality tyres will also give better mileage. Contrary to this relationship there is no correlation between tyre quality and fuel quality despite them exhibiting a strong correlation with mileage.
- b) Criteria for a mathematical function to be a logistic function

Continuity: The function should be continuous.

Differentiability: The function should be differentiable for all points in domain

Bounded Input: The function should have an upper and lower bound for its range, typically from 0 to 1

Monotonic: The function should be monotonically increasing or decreasing

Infinite input: The function should accept input from infinity to -infinity

- c) For sparse dataset 'leave-one-out' validation technique is used. In this technique each observation is chosen as the validation data and the rest n-1 observations as the training data. This is a more robust model compared to k-cross validation as it has number of folds=n. This method reduces the bias, randomness, mse and prevents overfitting. k-fold validation has some bias due to $(k-1)n/k$ observations in each training set. Leave one out has greater variance than k-fold due to more overlapping of training dataset. Leave one out is used for sparse dataset while k-fold is used for larger dataset.

- d) Linear Regression Line:

$$y = b_0 + b_1x$$

$$\text{sum of squared residuals } J = \sum (y_i - \hat{y}_i) = \sum (y_i - (\hat{b}_0 + \hat{b}_1x))^2$$

We have to minimize this sum by partial differentiating wrt to b_0 and b_1

$$\frac{\partial J}{\partial b_0} = \sum 2(y_i - (\hat{b}_0 + \hat{b}_1x_i))(-1)$$

$$= -2\sum (y_i - (\hat{b}_0 + \hat{b}_1x_i))$$

$$-2\sum (y_i - (\hat{b}_0 + \hat{b}_1x_i)) = 0$$

$$\sum (y_i - (\hat{b}_0 + \hat{b}_1x_i)) = 0$$

$$n\hat{b}_0 = \sum y_i - \sum \hat{b}_1x_i$$

$$\hat{b}_0 = \frac{\sum y_i}{n} - \frac{\hat{b}_1 \sum x_i}{n}$$

$$\hat{b}_0 = \bar{y} - \hat{b}_1\bar{x}$$

$$\begin{aligned}\frac{\partial J}{\partial b_1} &= \sum 2(y_i - (\widehat{b_0} + \widehat{b_1}x_i))(-x_i) \\ -2\sum(x_i)(y_i - (\widehat{b_0} + \widehat{b_1}x_i)) &= 0 \\ \sum(x_i)(y_i - (\bar{y} - \widehat{b_1}\bar{x} + \widehat{b_1}x_i)) &= 0 \\ \sum(x_i)(y_i - \bar{y}) &= \sum\widehat{b_1}x_i(x_i - \bar{x}) \\ \widehat{b_1} &= \frac{\sum x_i(y_i - \bar{y})}{\sum x_i(x_i - \bar{x})} \\ \widehat{b_1} &= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}\end{aligned}$$

e) A) α, β, σ

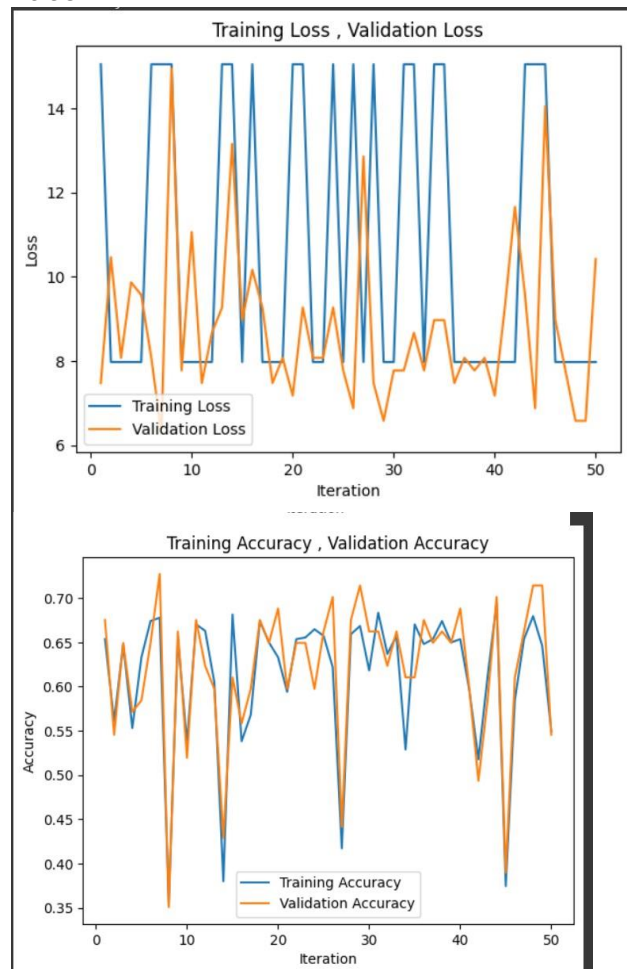
f) D) $y = \alpha + \beta_1x + \beta_2x^2 + \epsilon$ $\beta_2 > 0$

Section B(Bonus)

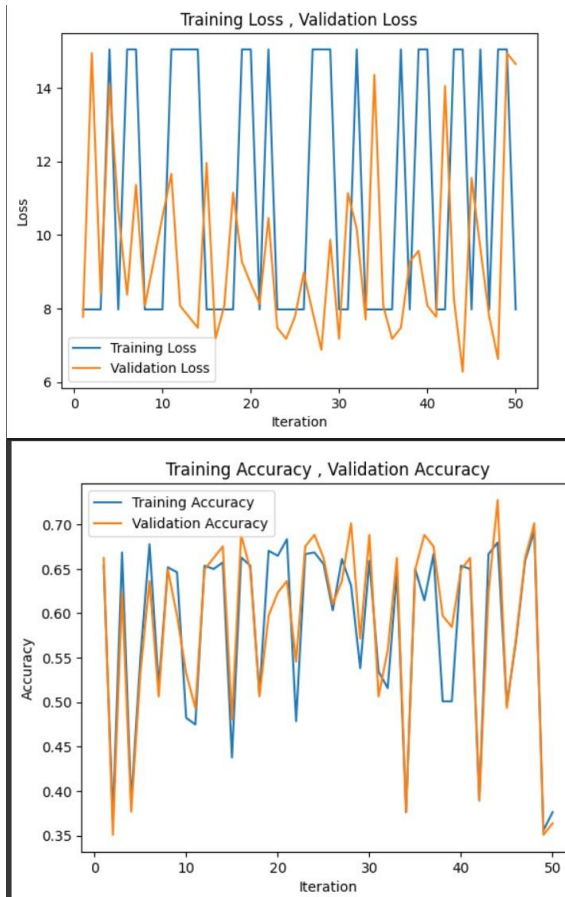
a) test loss: 6.793712945087064
test accuracy: 0.6883116883116883

The model doesn't converge as the training and validation loss doesn't stabilize

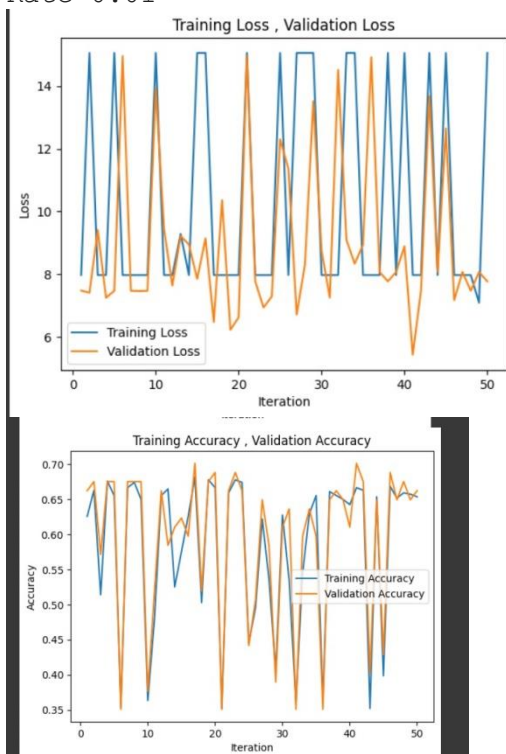
b) rate=1



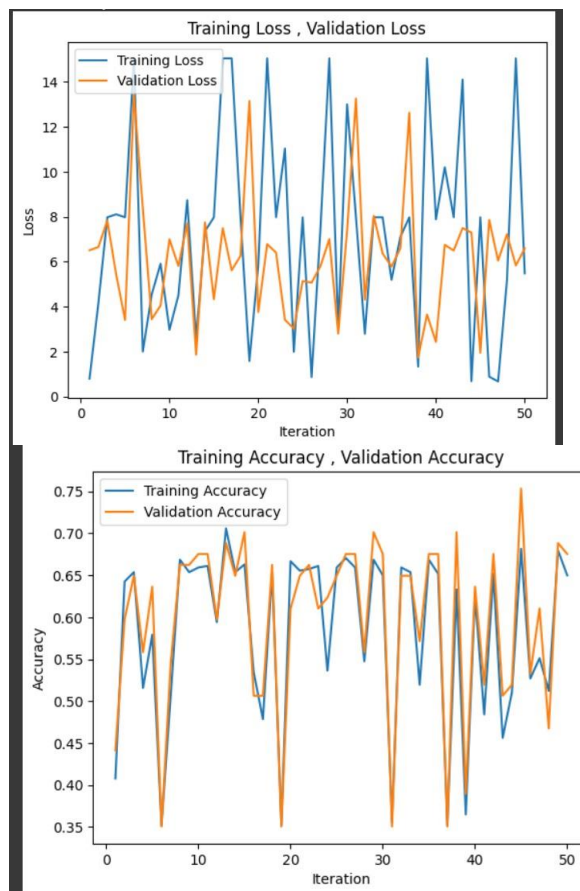
Rate=0.1



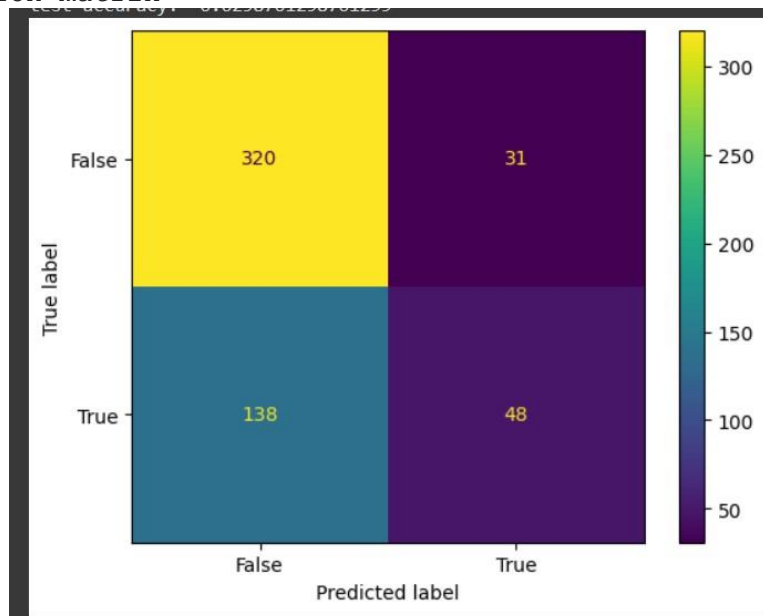
Rate=0.01



Rate=0.001

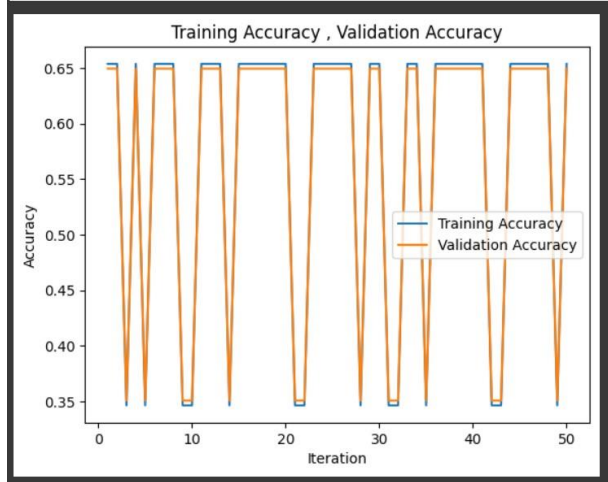
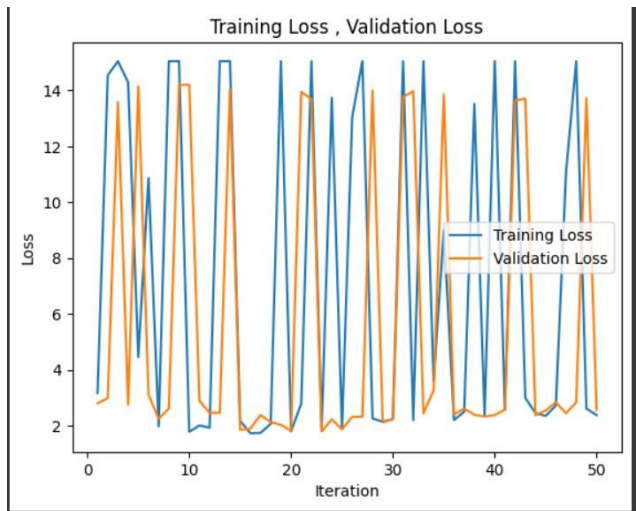


c) confusion matrix

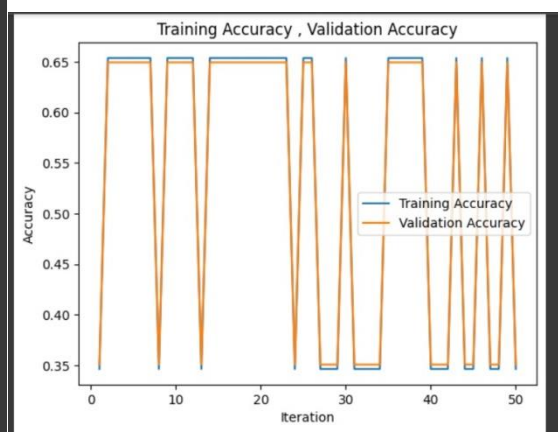
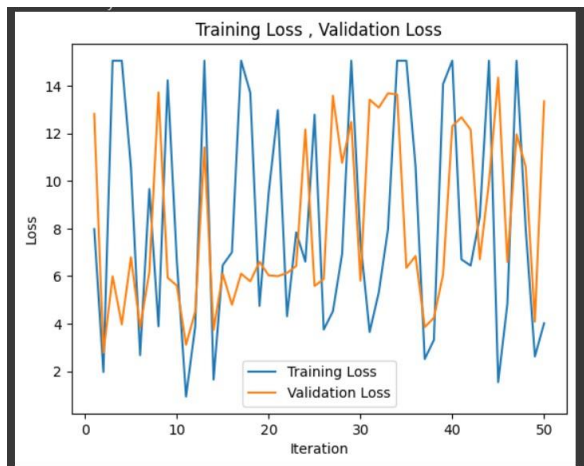


Precision=0.607
Accuracy=0.68
Recall=0.258
F1=0.37

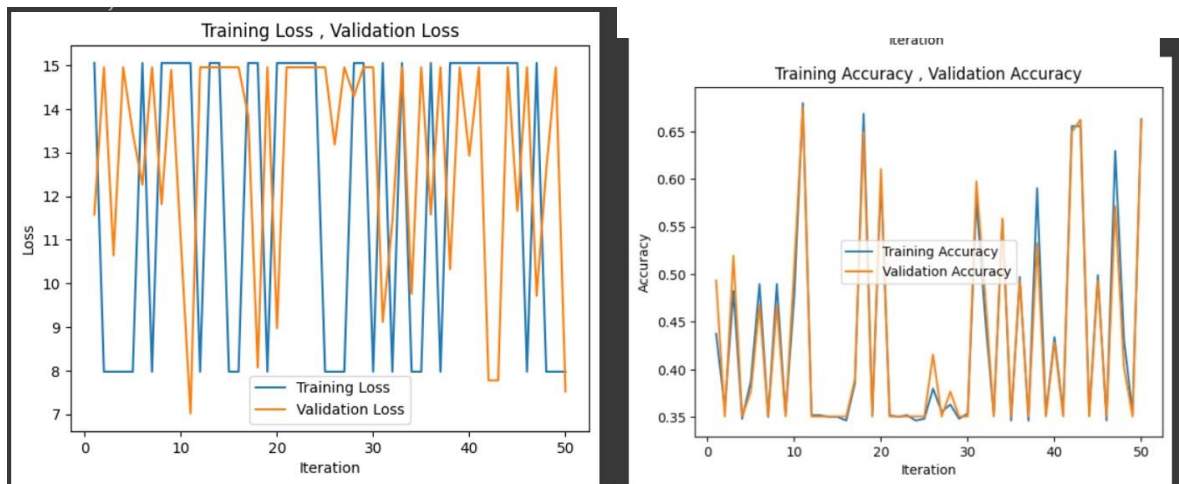
d) 11



L2

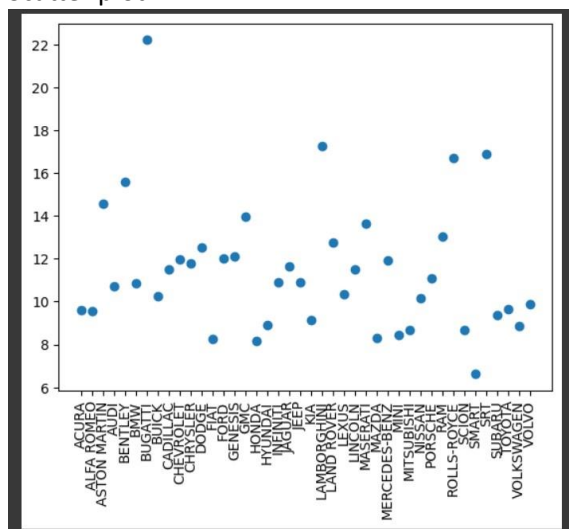


e) tanh

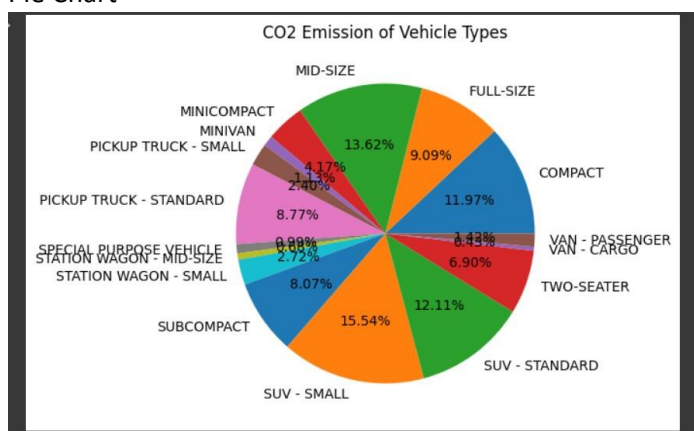


Section C

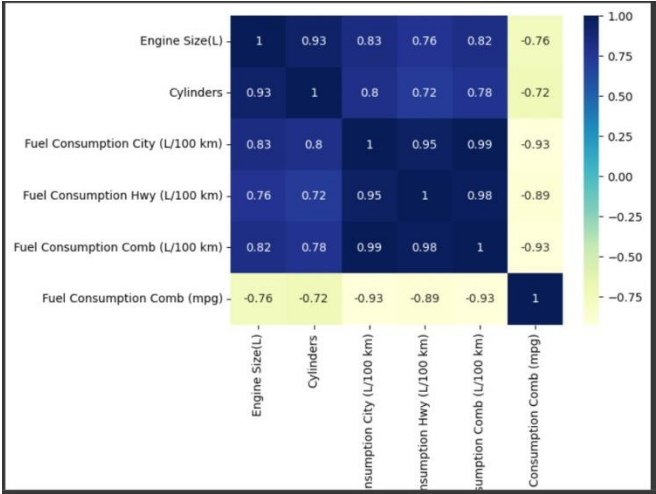
a) Scatter plot



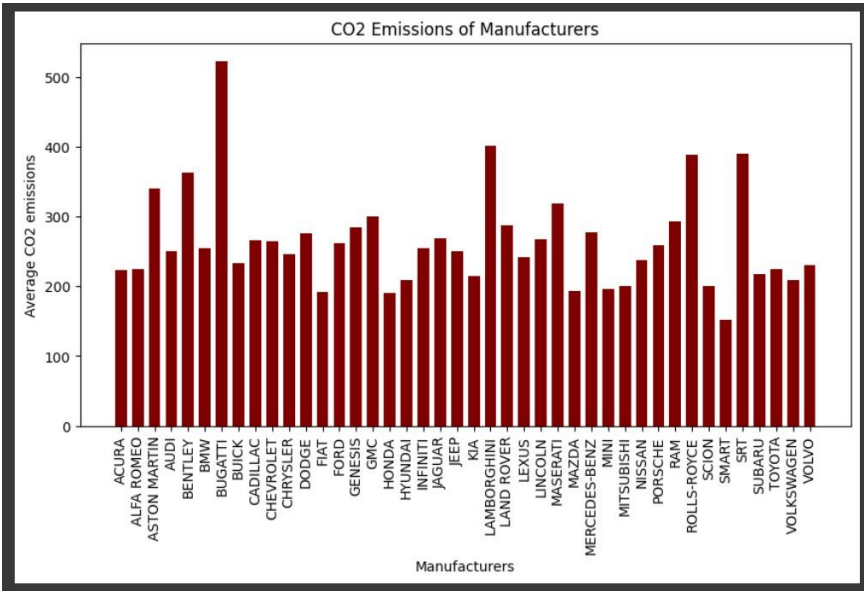
Pie Chart



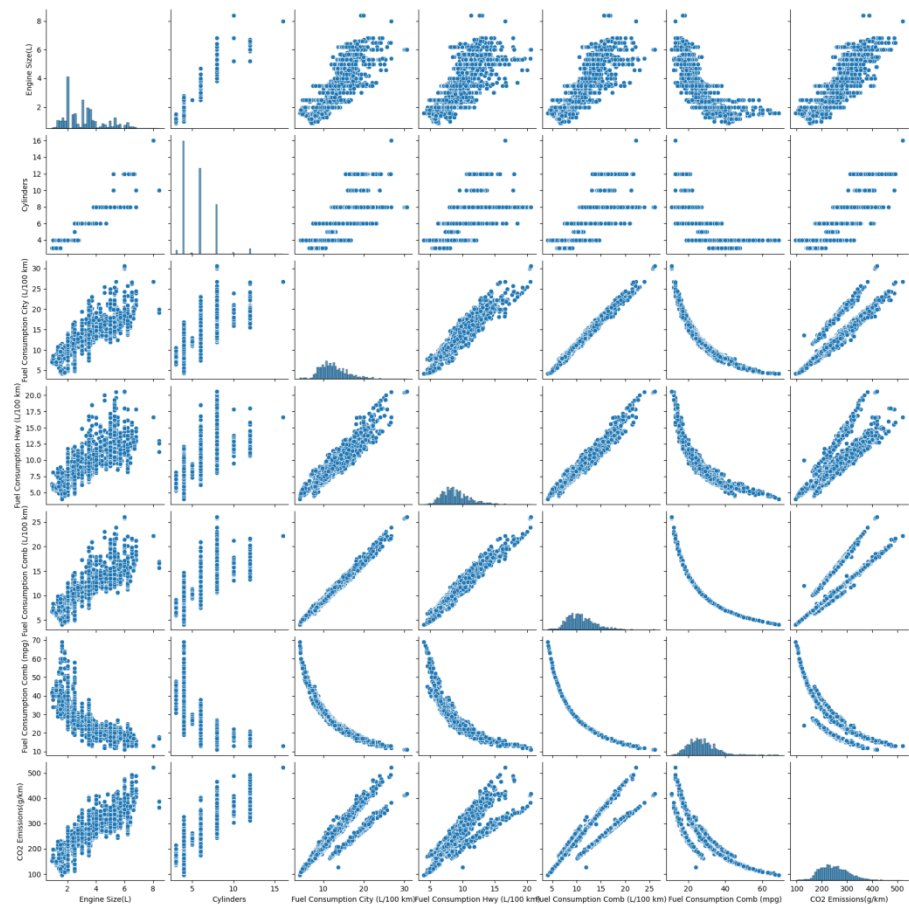
Heatmap



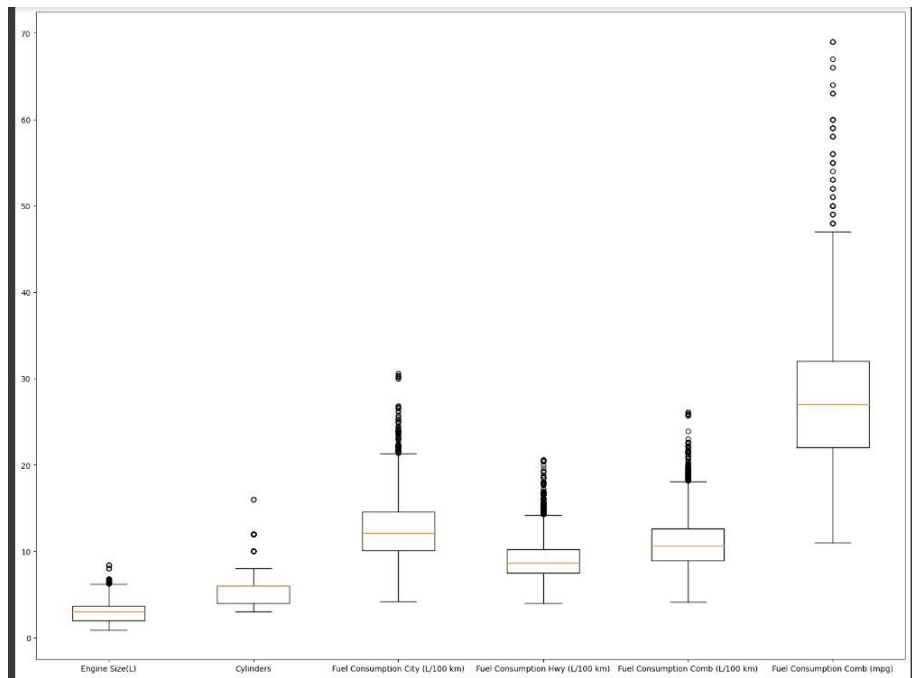
Bar Graph



Pair plot



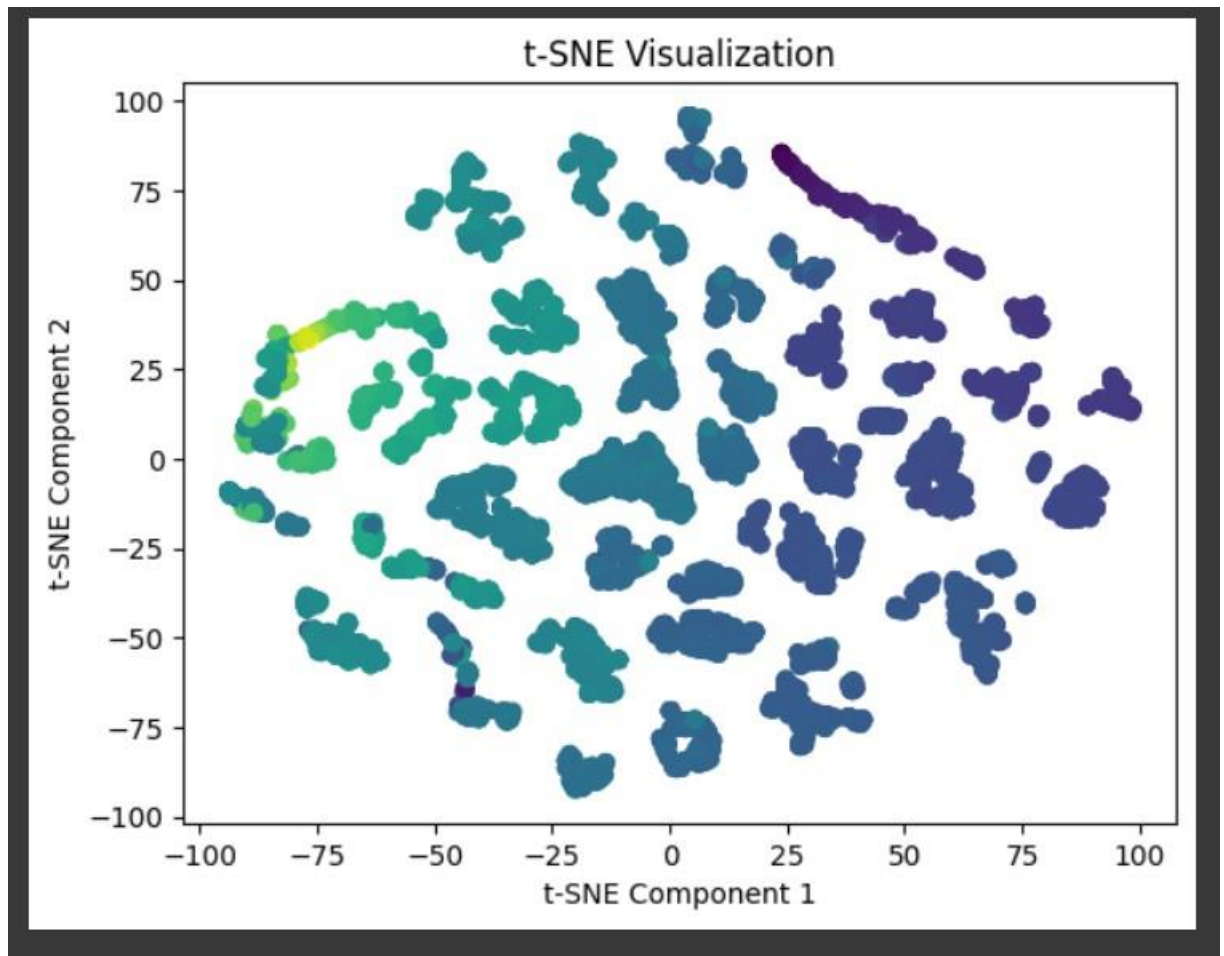
Box plot



Insights

1. Analyzing the dataset reveals that smaller vehicle classes (e.g., subcompact and compact) tend to have better fuel efficiency in both city and highway conditions compared to larger vehicle classes (e.g., SUVs and trucks).
2. There is a clear correlation between engine size (measured in liters) and fuel consumption. Vehicles with larger engine sizes tend to have higher fuel consumption, especially in city driving conditions.
3. Vehicles with automatic transmissions generally have slightly higher fuel consumption compared to those with manual transmissions.
4. Fuel consumption directly affects the CO₂ emission. Higher consumption leads to higher emission.
5. Generally, vehicles with higher MPG tend to have lower CO₂ emissions.
6. Bugatti is an outlier as it has a very high CO₂ emission due to its powerful engine and higher fuel consumption.

b)



The data is not separable due to large number of clusters

c) Linear Regression with label encoding

```
train data
mean squared error: 272.4546028616039
rmse: 16.506198922271714
r2 score: 0.9173457220175019
r2 adjusted 0.917205558751464
mae: 10.92201183067301

test data
mean squared error: 286.3092754463559
rmse: 16.920675974864476
r2 score: 0.9142804706918566
r2 adjusted 0.9136957535751571
mae 11.139985999069903
```

d) PCA on different components

index	no of components	mse_train	mse_test	rmse_train	rmse_test	r2_train	r2_test	r2_adjusted_train	r2_adjusted_test	mae_train	mae_test
0	4	272.4546	286.3093	16.5062	16.92068	0.917346	0.91428	0.917206	0.913696	10.92201	11.13999

1	6	272.4546	286.3093	16.5062	16.92068	0.917346	0.91428	0.917206	0.913696	10.92201	11.13999
2	8	272.4546	286.3093	16.5062	16.92068	0.917346	0.91428	0.917206	0.913696	10.92201	11.13999
3	10	272.4546	286.3093	16.5062	16.92068	0.917346	0.91428	0.917206	0.913696	10.92201	11.13999

e) Linear Regression with one hot encoding

- Label-encoding

```
train data
mean squared error: 272.4546028616039
rmse: 16.506198922271714
r2 score: 0.9173457220175019
r2 adjusted 0.917205558751464
mae: 10.92201183067301

test data
mean squared error: 286.3092754463559
rmse: 16.920675974864476
r2 score: 0.9142804706918566
r2 adjusted 0.9136957535751571
mae 11.139985999069903
```

One hot encoding

```
train data
mean squared error: 8.534904307079321
rmse: 2.921455854035676
r2 score: 0.9975033561715226
r2 adjusted 0.9960819141618449
mae: 1.8711311402317634

test data
mean squared error: 4.392014384390629e+20
rmse: 20957133354.518288
r2 score: -1.2768866668037675e+17
r2 adjusted 2.8256142731669578e+17
mae 4254462400.7264833
```

- On train data the one-hot encoding performs better but on the test data the label encoding performs much better.

f) PCA with one hot encoding

in de	no of compone	mse_train	mse_test	rmse_train	rmse_test	r2_train	r2_test	r2_adjusted_train	r2_adjusted_test	mae_train	mae_test
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x	nts		t	n						n	
0	4	335. 4253	342. 2445	18.31 462	18.4 9985	0.90 1881	0.90 05	0.901831	0.90029 7	11.9 5959	11.9 6072
1	6	325. 1359	332. 5278	18.03 152	18.2 3535	0.90 4891	0.90 3324	0.90481	0.90299 6	11.9 6451	11.9 4435
2	8	309. 7489	322. 2432	17.59 969	17.9 5113	0.90 9392	0.90 6315	0.909284	0.90586 8	11.5 3319	11.7 5756
3	10	301. 0083	310. 9263	17.34 959	17.6 331	0.91 1949	0.90 9605	0.911814	0.90905	11.5 3785	11.7 4879
4	12	287. 9775	290. 1298	16.96 99	17.0 332	0.91 576	0.91 5651	0.915603	0.91501 7	11.3 1824	11.5 4672

g) L1 and L2 regularization

l1 regularization

train data

mean squared error: 276.0006730208568

rmse: 16.6132679813713

r2 score: 0.9164177847397781

r2 adjusted 0.9162760478985703

mae: 11.090616160367098

test data

mean squared error: 291.2262650056135

rmse: 17.0653527653434

r2 score: 0.9121284033614612

r2 adjusted 0.911529006385755

mae 11.492813366693914

l2 regularization

train data

mean squared error: 278.6699517969926

rmse: 16.69341043037619

r2 score: 0.9151057311707361

r2 adjusted 0.9149617693785889

mae: 11.10470337885928

test data

mean squared error: 261.3856458726752
rmse: 16.167425455918305
r2 score: 0.9229307341153539
r2 adjusted 0.9224050228883099
mae 10.854228473092643

h) SGD Regression

train data

mean squared error: 276.8278071322528
rmse: 16.638143139552948
r2 score: 0.917181918893925
r2 adjusted 0.9170414778542335
mae: 11.144264509272686

test data

mean squared error: 270.12461956275826
rmse: 16.43546834023169
r2 score: 0.9144184320231268
r2 adjusted 0.9138346559796283
mae 11.03469746732977